

PH331 : Computational Physics Lab

Chapter III

Projectile Motion: Solving Differential Equations

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1 Problem 1: Free projectile

Solve the two dimensional trajectory of a projectile fired from a cannon shell located at the origin using the Euler method. Assume the initial projected speed is 700 ms^{-1} , and using different firing angles starting from 20 degrees to 60 degrees with an interval of 5 degrees. Neglect the effects of the air resistance. Plot the trajectories of the projectile for different firing angles. Also plot the range of the projectile against the firing angles and show numerically that the maximum range of the projectile corresponds to a firing angle of 45 degrees. Compare the numerical results with the analytical solutions as far as the range and the duration of the projectile.

1.1 Program Code:

```
#include <stdio.h>
#include <math.h>

int main()
{
    double v=700.0,v_x,v_y,x,y,dt=0.05,t=0,g=-9.8;
    double max=0,tt=0,range;
    FILE *f,*ran;
    int theta,i,th_m;
    char file[20];
    ran=fopen("normal.txt","w");
    printf("\nFiring Angle\tTheoretical Range,Time\t\tNumerical Range,Time\n\n");
    for(theta=20;theta<=60;theta+=5)
    {
        v_x=v*cos(theta*M_PI/180.0);
        v_y=v*sin(theta*M_PI/180.0);
        y=x=t=0.0; i=0;
        tt = (2.0*v_y)/(-1*g);
        range = v*v*sin(2*theta*M_PI/180.0)/(-1*g);
        sprintf(file,"a%d.txt",theta);
        f=fopen(file,"ab+");
```

```

do{
    if(i%20==0)
        fprintf(f, "%lf\t%lf\t%lf\n",t,y,x);
    v_y=v_y+g*dt;
    y=y+v_y*dt;
    x=x+v_x*dt;
    t=t+dt;
    ++i;
}while(y>=0);
fprintf(f, "%lf\t%lf\t%lf\n",t,y,x);
printf("%d\t\t%lf, %.8lf\t%lf, %lf\n",theta,range,tt,x,t);
fclose(f);
if(x>max)
{
    th_m=theta;
    max=x;
}
fprintf(ran, "%d\t%lf\t%lf\n",theta,x,range);
}
printf("\nMaximum range is for firing angle =%d deg\n",th_m);
fclose(ran);
}

```

1.2 Output:

```
[student@localhost SC12B156]$ ./prog31
```

Firing Angle	Theoretical Range,Time	Numerical Range,Time
20	32139.380484, 48.86002048	32132.789168, 48.850000
25	38302.222156, 60.37403739	38286.972463, 60.350000
30	43301.270189, 71.42857143	43283.949681, 71.400000
35	46984.631039, 81.93949091	46961.986699, 81.900000
40	49240.387651, 91.82680138	49226.015915, 91.800000
45	50000.000000, 101.01525446	49992.449430, 101.000000
50	49240.387651, 109.43492045	49224.675150, 109.400000
55	46984.631039, 117.02172061	46975.910137, 117.000000
60	43301.270189, 123.71791483	43295.000000, 123.700000

Maximum range is for firing angle =45 deg

```
[student@localhost SC12B156]$ gnuplot
```

Following are the observations:

- The trajectory of the projectiles is symmetric about the highest point of flight.
- The range increases upto 45° for the same speed and then decreases.
- The time of flight and maximum height increase with firing angle.

From the data from the above program the following plots were made using **gnuplot**:

Figure 1: Projectile Trajectory without air resistance

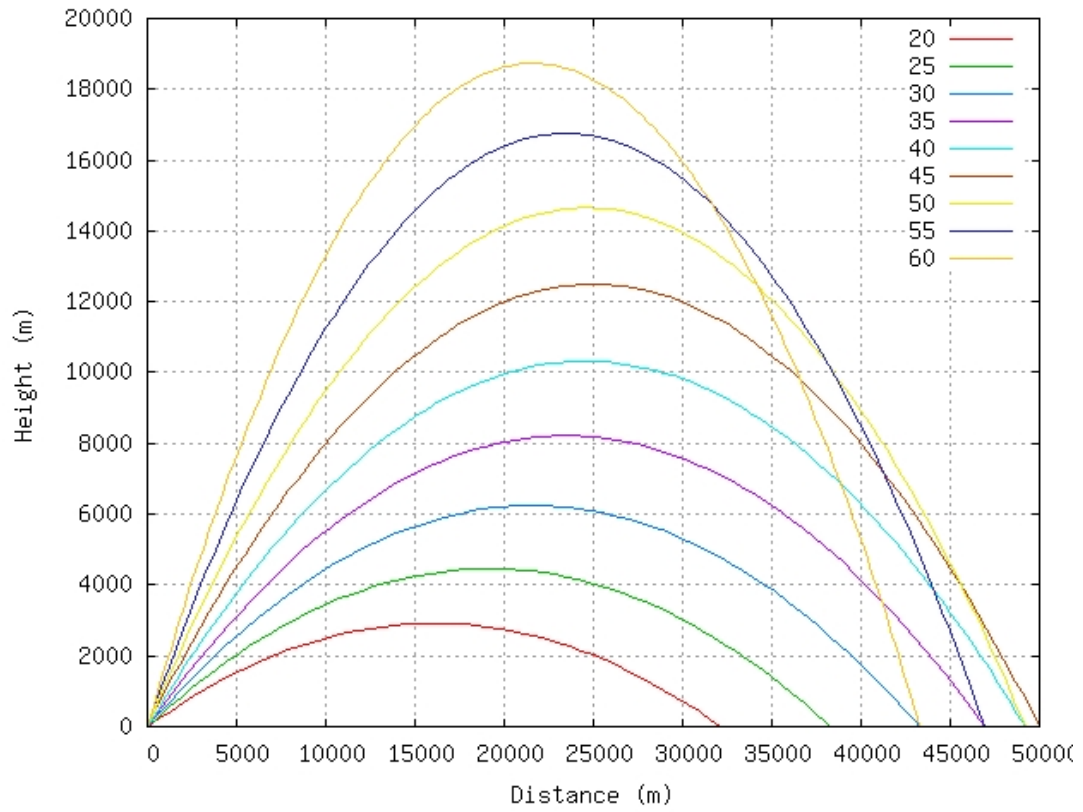
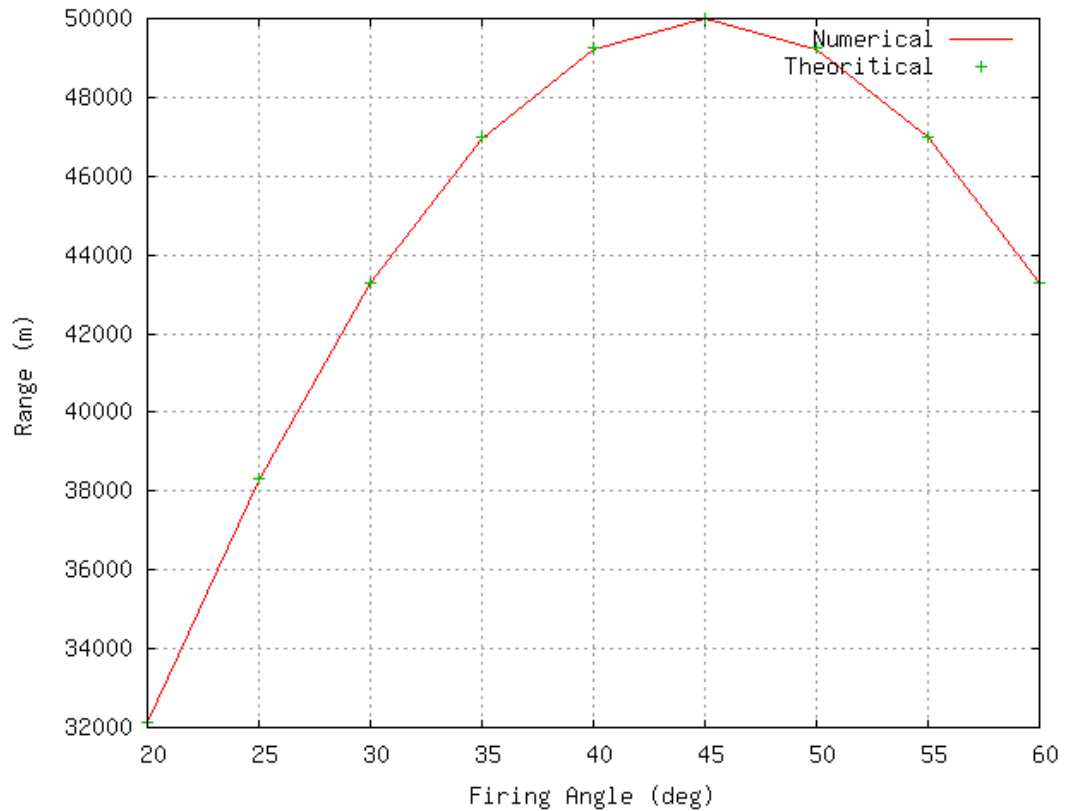


Figure 2: Range vs Firing Angle



1.3 Discussion:

For a two dimensional projectile motion without air resistance the equations of motions can be very easily worked out by directly applying the Equations of Motion or solving the trivial differential equation(s) obtained by Newton's laws of Motion ($\frac{d\mathbf{v}}{dt} = \mathbf{g}$). The following equations describe the trajectory:

- Trajectory: (Parabolic) $y = x \tan(\theta) - \frac{g x^2 \sec^2(\theta)}{2v^2}$
- The Horizontal Range is given by: $R = \frac{v^2 \sin(2\theta)}{g}$
- The flight duration is given by: $t = \frac{2v \sin(\theta)}{g}$

Thus as $\sin(\theta)$ increases in 0° to 90° and decreases from 90° to 180° , according to the equations above the range increases upto 45° and decreases for higher angles. Also the maximum height increases with firing angle (more vertical velocity component) which in turn leads to increase in duration of flight. This is also in agreement of our numerical result.

2 Problem 2: Air resistance

Extend the above problem to include the effect of the air resistance by assuming a constant value of $B/m = 4 \times 10^{-5} m^{-1}$. Plot the trajectories for a given firing angle for the case of no air resistance and with air resistance. Repeat this for different firing angles. Also plot the range of the projectile against the different firing angles and find numerically the firing angle which corresponds to the maximum range of the projectile.

2.1 Program Code:

```
#include <stdio.h>
#include <math.h>

int main()
{
    double v=700.0, v_x, v_y, x, y, dt=0.005, t=0, g=-9.8, b=4e-5, a=0, max=0;
    FILE *f, *ran;
    int theta, i, th_m;
    char file[20];
    ran=fopen("res.txt", "w");
    for(theta=20; theta<=60; theta+=5)
    {
        v=700.0;
        v_x=v*cos(theta*M_PI/180.0);
        v_y=v*sin(theta*M_PI/180.0);
        x=y=t=0.0;
        i=0;
        sprintf(file, "b%d.txt", theta);
        f=fopen(file, "wb");
        do{
            if(i%200==0)
                fprintf(f, "%lf\t%lf\t%lf\n", t, y, x);
            a=g-b*v*v_y;
            v_y=v_y+a*dt;
```

```

y=y+v_y*dt;

v_x=v_x-b*v*v_x*dt;
x=x+v_x*dt;
t=t+dt;
v=sqrt(v_x*v_x+v_y*v_y);
++i;
}while(y>=0);
fprintf(f,"%lf\t%lf\t%lf\n",t,y,x);
if(x>max)
{
    th_m=theta;
    max=x;
}
fprintf(ran,"%d\t%lf\n",theta,x);
fclose(f);
}
printf("Maximum range is for firing angle =%d deg\n",th_m);
fclose(ran);
}

```

2.2 Output:

```

[student@localhost SC12B156]$ ./prog32
Maximum range is for firing angle = 40 deg
[student@localhost SC12B156]$ gnuplot

```

From the data from the above program the following plots were made using **gnuplot**:

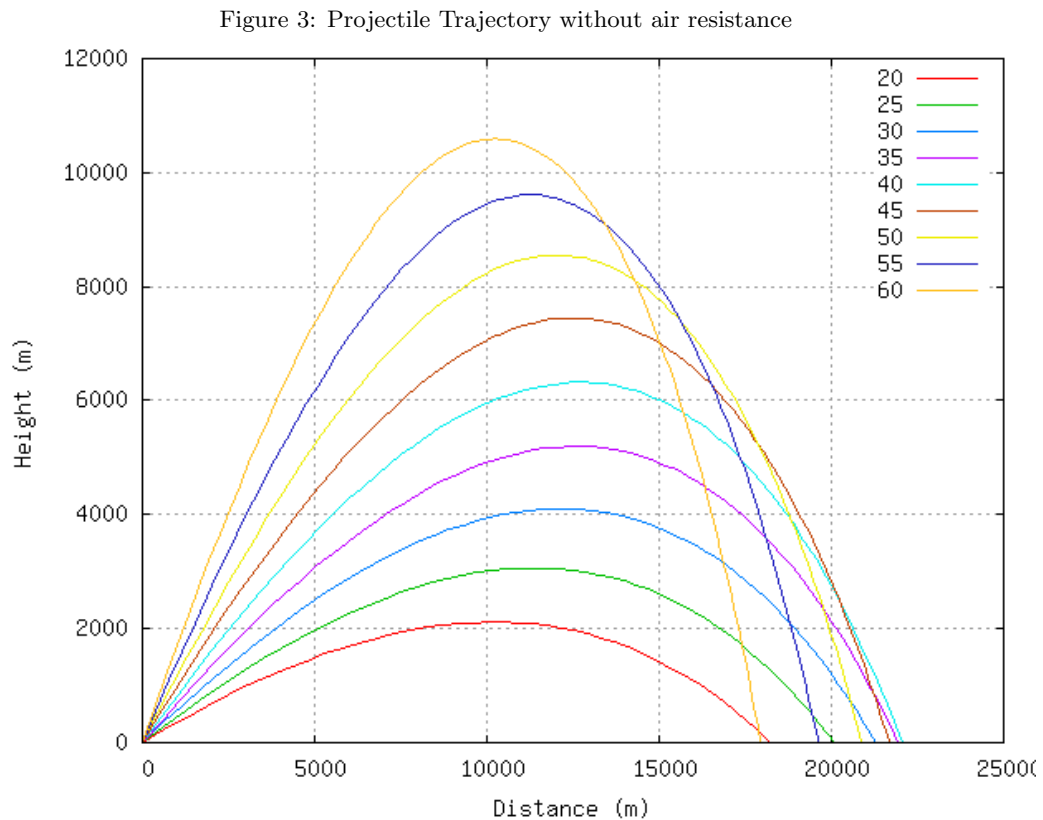
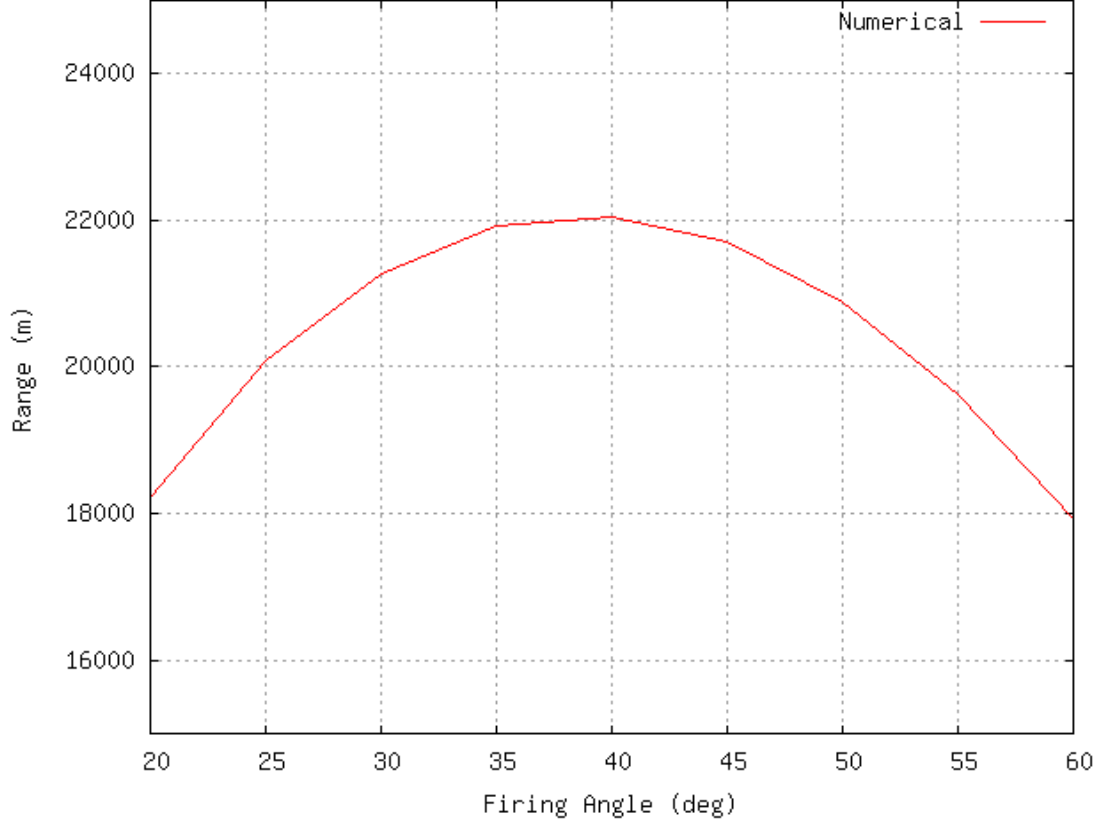


Figure 4: Range vs Firing Angle



Following are the observations:

- The trajectory of the projectiles is no longer symmetric about the highest point of flight.
- The range is drastically reduced with the maximum range corresponds approximately to a firing angle of 40° .

2.3 Discussion

For a two dimensional projectile motion with air resistance the equations of motions are quite complicated compared to the previous case. The differential equation obtained by Newton's laws of Motion is $(\frac{d\mathbf{v}}{dt} = \mathbf{g} - B \cdot \mathbf{v} \cdot \mathbf{v})$. It can be split into equations for Distance (x) and Height (y) axes as follows:

- y -axis:
$$\frac{d^2y}{dt^2} = g - B \cdot v \cdot \frac{dy}{dt}$$
- x -axis:
$$\frac{d^2x}{dt^2} = -B \cdot v \cdot \frac{dx}{dt}$$

Here, $B = \frac{1}{2}\rho c_D A$ and $v = \sqrt{(\frac{dy}{dt})^2 + (\frac{dx}{dt})^2}$

The air density (ρ) is assumed to be constant, c_D is the air drag coefficient and A is the area of the body (which do not concern us as B is constant). Note that the drag always tries to nullify the velocity.

The differential equations were solved computationally using Euler's Method. We find the velocity decays exponentially due to the damping effect of air resistance. This leads to nearly **half the range** obtained in the previous problem for the same initial conditions. We also see that the maximum range occurs at 40° instead of 45° from Fig.4. Thus to model real life dynamics, the drag experienced by the

body cannot be neglected.

Also the ratio of range with variable air resistance to the normal range decreases with increasing firing angle as duration of flight increases for higher angles and thus more drag is experienced.

3 Problem 3: Inhomogeneous air resistance

Now consider the air resistance is not uniform. Consider the air density varies as $\exp(-y/y_0)$, where $y_0 = 10^4$ meters. Now plot the trajectories for a given firing angle without air resistance, with constant air resistance and with varying air resistance as given above. Repeat this for different firing angles. Also plot the range of the projectile against the different firing angles and find numerically the firing angle which corresponds to the maximum range of the projectile.

3.1 Program Code:

```
#include <stdio.h>
#include <math.h>

int main()
{
    double v=700.0,v_x,v_y,x,y,dt=0.005,t=0,g=-9.8,b=4e-5,a=0,y0=1e4,max=0;
    FILE *f,*ran;
    int theta,i,th_m;
    char file[20];
    ran=fopen("varres.txt","w");
    for(theta=20;theta<=60;theta+=5)
    {
        v=700.0;
        v_x=v*cos(theta*M_PI/180.0);
        v_y=v*sin(theta*M_PI/180.0);
        y=x=t=0.0;
        i=0;
        sprintf(file,"c%d.txt",theta);
        f=fopen(file,"wb");
        do{
            if(i%200==0)
            {
                fprintf(f,"%lf\t%lf\t%lf\n",t,y,x);
            }
            a=g-b*exp(-1*y/y0)*v*v_y;
            v_y=v_y+a*dt;
            y=y+v_y*dt;

            v_x=v_x-b*exp(-1*y/y0)*v*v_x*dt;
            x=x+v_x*dt;
            t=t+dt;
            v=sqrt(v_x*v_x+v_y*v_y);
            ++i;
        }while(y>=0);
        fprintf(f,"%lf\t%lf\t%lf\n",t,y,x);
        fclose(f);
    }
}
```

```

    if(x>max)
    {
        th_m=theta;
        max=x;
    }
    fprintf(ran,"%d\t%lf\n",theta,x);
}
printf("Maximum range is for firing angle =%d deg\n",th_m);
fclose(ran);
}

```

3.2 Output:

```

[student@localhost SC12B156]$ ./prog3
Maximum range is for firing angle = 45 deg
[student@localhost SC12B156]$ gnuplot

```

From the data from the above program the following plots were made using **gnuplot**:

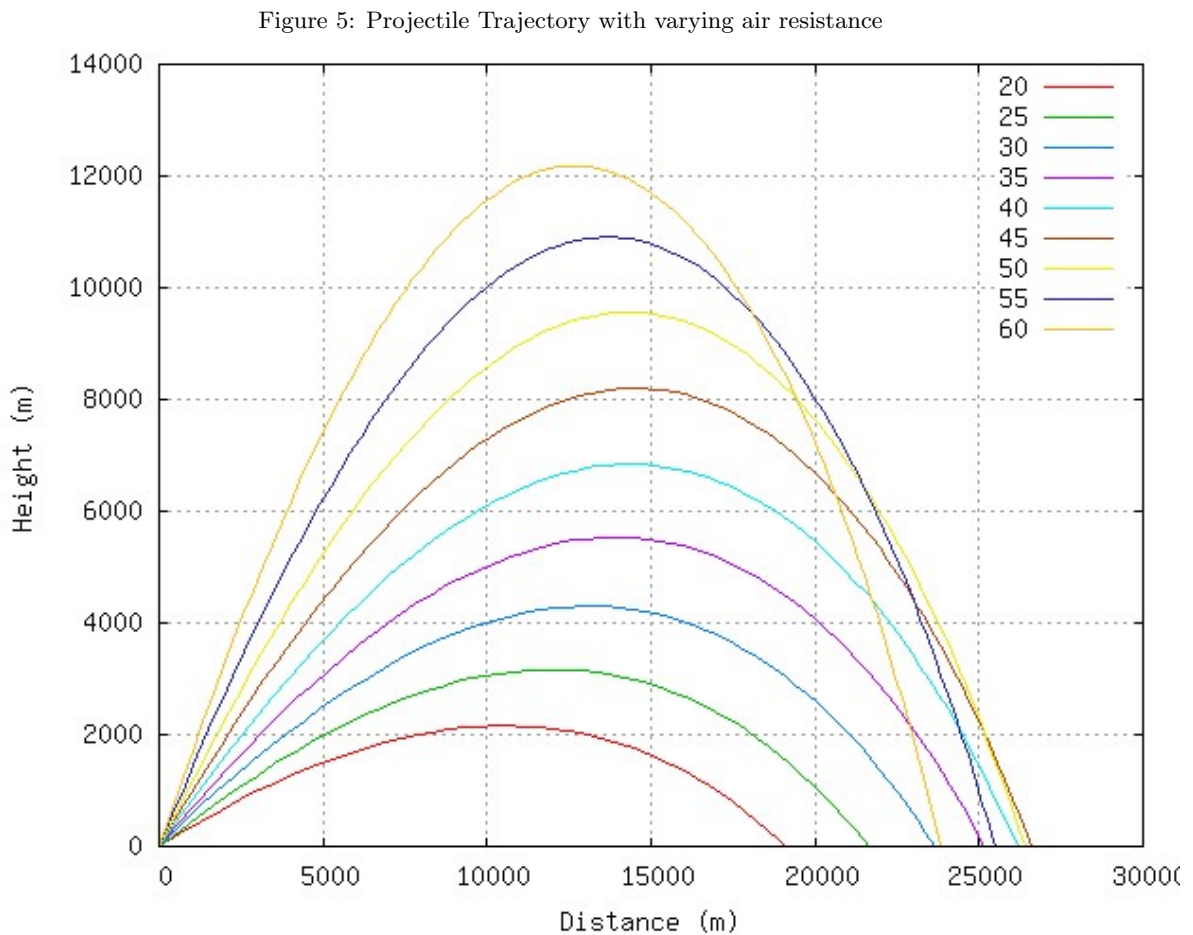
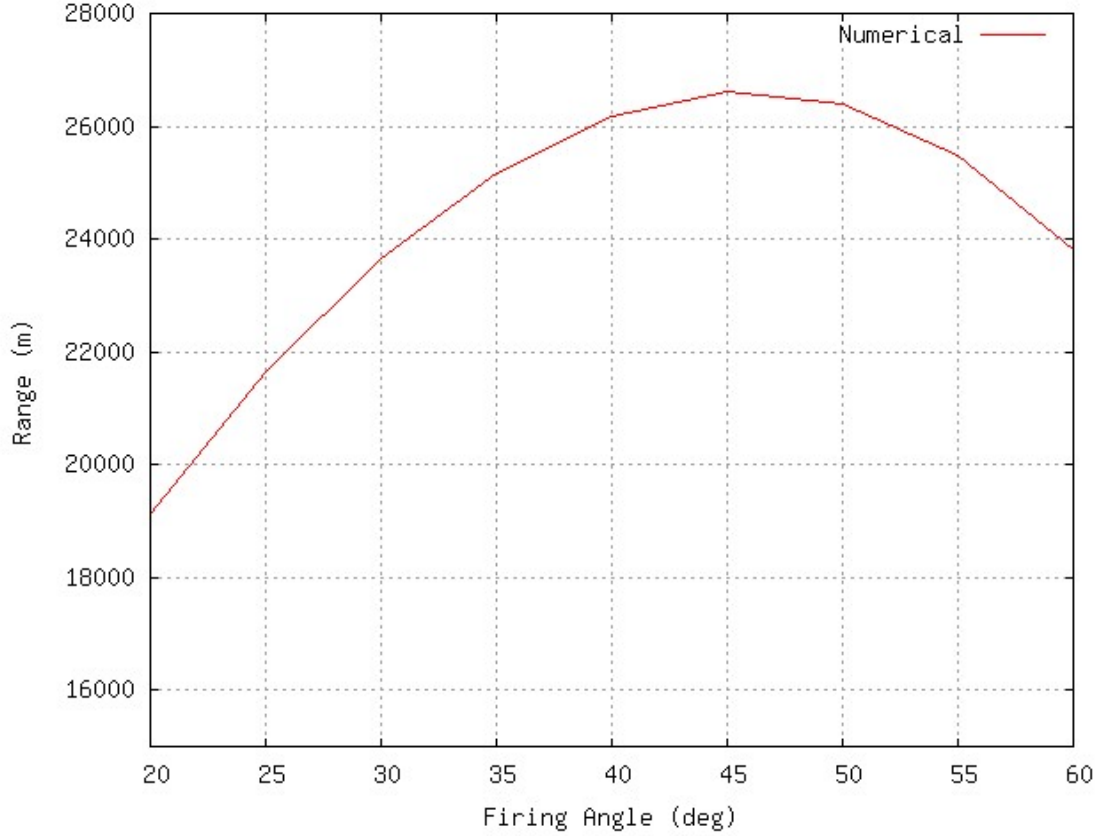


Figure 6: Range vs Firing Angle



Following are the observations:

- The trajectory of the projectiles is no longer symmetric about the highest point of flight. The
- The range is slightly more than above case but still does not follow a "simple" relationship with the firing angle.
- The maximum range corresponds approximately to a firing angle of 45° .

3.3 Discussion

For a 2D projectile motion with variable air resistance the equations of motions are even more complex. The differential equation obtained by Newton's laws of Motion is $(\frac{d\mathbf{v}}{dt} = \mathbf{g} - B(y) \cdot \mathbf{v} \cdot \mathbf{v})$. For different axes - Distance (x) and Height (y) we have:

- y -axis:
$$\frac{d^2y}{dt^2} = g - B(y) \cdot v \cdot \frac{dy}{dt}$$
- x -axis:
$$\frac{d^2x}{dt^2} = -B(y) \cdot v \cdot \frac{dx}{dt}$$

Here, $B = \frac{1}{2} A c_D \rho_0 e^{-\frac{y}{y_0}}$, $y_0 = 10^4$ meters and $v = \sqrt{(\frac{dy}{dt})^2 + (\frac{dx}{dt})^2}$

The air density (ρ) is assumed to vary exponentially being max. at ground ($y=0$), c_D is the air drag coefficient and A is the area of the body (which do not concern us as B_0 is constant).

We reduce: $B = B_0 e^{-\frac{y}{y_0}}$

The differential equations were solved computationally using Euler's Method. We find the velocity decays almost exponentially due to the damping effect of air resistance. This leads to marginally

more than **half the range** obtained in the first problem. The range is greater than the constant air resistance (or density) case as drag decreases with height. We also find the maximum range occurs at again at 45° from Fig. 6. Also the ratio of range with variable air resistance to the normal range increases with firing angle as drag reduces appreciably with height (even though duration of flight increases).

The comparison of the above 3 cases for firing angle of 45° is shown below:

